

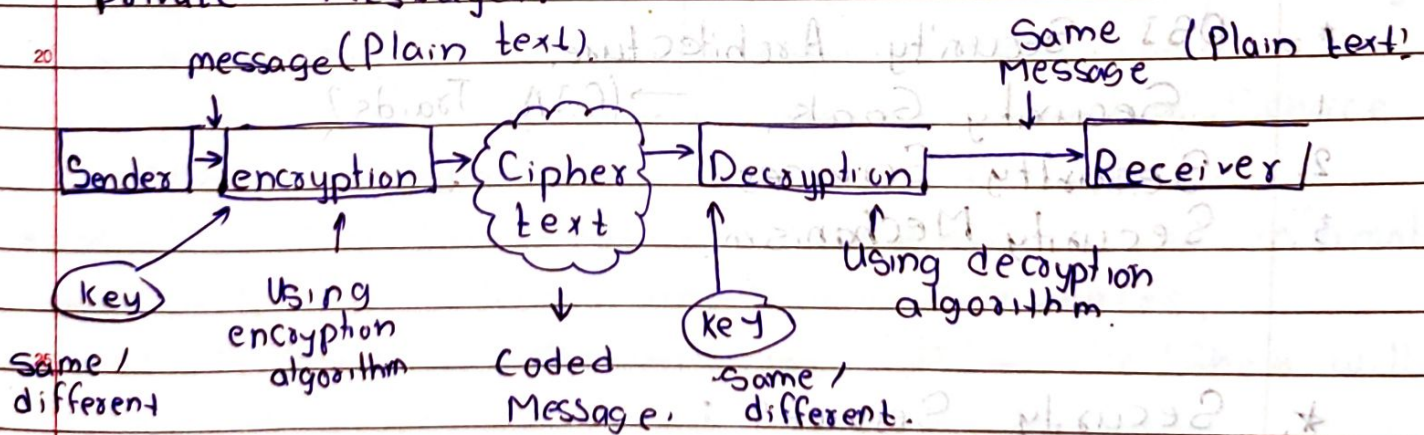
* **Computer Security** : The protection afforded to an automation information system in order to attain the applicable objectives of preserving integrity, availability and confidentiality of information system resources such as hardware, software data or information and telecommunication.

There are more two additional triads they are :

- Authenticity
- Accountability

* **Cryptography** : The art of protecting information by transforming it into unreadable format is cryptography.

More generally cryptography is about constructing and analysing protocols that prevent third-parties or the public from reading private messages.



* **Types of Cryptography**

- Symmetric

- Asymmetric

- If keys are same it is Symmetric key cryptography (Private Key)
- If keys are different it is asymmetric key cryptography. (Public)

Symmetric Cryptography

• Conventional encryption technique private key cryptography.

- One Key
- Faster
- less Complex.
- Used for bulk data.

• Commonly used algorithms are DES, AES

Asymmetric Cryptography

• It is also called public key cryptography.

- Two different key
- Slower.
- More Complex.
- Used for exchange of secret keys.

• Commonly used algorithms are RSA, Diffie Hellman etc.

Algo used.

* OSI Security Architecture

1. Security Goals, → (CIA Triad)
2. Security Services
3. Security Mechanism

* Security Services :

1. Data Confidentiality - Protect data from attackers.
2. Data Integrity - Protecting data from modification
3. Authentication - Verifying actual person is ^{the validity of something} trusted or not
4. Non repudiation - Assurance that someone cannot deny ^{the validity of something}
5. Access Control - To whom the access should be given can be decided.

* Security Mechanism:

1. Encipherment: Use of mathematical algorithm to transform data to unreadable form.
2. Digital Signature: Sender can electronically sign the data and receiver can electronically verify the signature to check whether the message is modified or not.
3. Data Integrity: It provides a check value by the sender.
4. Authentication exchange: Two entities exchange some message to prove their identity to each other. ex. bluetooth handshake. Extra bits are added to actual data.
5. Traffic Padding: Selecting and continuously changing different available routes to avoid eaves dropping.
6. Routing Control: It is method to that a user has access right of the data.
7. Access Control: Means selecting a third trusted party to control the communication between two entities.
8. Notarization: This can be done to prevent repudiation.

* Security Attacks.

Security Attacks.

Passive

- Release of message content
- Traffic Padding (Encryption)

Active

- Mosquerade
- Modification of message
- Reply
- Denial of Service

* Passive Attacks: It attempts to learn or make use of information but does not affect the System's resources.

- The attacker can only see the data. He/She will not modify it.
- We can prevent it by using better encryption techniques.
- Two types of passive attacks are.
 - 1] Release of message content.
 - 2] Passive attacks are difficult to detect.

* Active Attacks: Attacker can see and modify the message.

• Four types of active attacks are

1) Masquerade: When one entity pretends to be another.

2) Modification of message: Msg is altered or delayed.

3) Replay: Retransmission of message is called replay.

4) DOS: Overloading the server to degrade the performance and doesn't allow server to fulfill any request.

* Classical Encryption Techniques:

• Symmetric encryption also referred as conventional encryption which are of two types.

1. Substitution.

2. Transposition.

* Substitution: Substitution is one in which letter of plain text are replaced by another letter, numbers or symbols.

A - P

B - 1

C - Z

D - @

E - \$

CAB → ZP1

Mono Alphabetical
Polyalphabetical

* Transposition: Performing some sort of permutation on the plain text letter.

i.e. it reorders the symbols or letters.

Name → mane, aenm, mnea

Types of transposition

- Keyless
- Keyed.

ex.

- Rail Fence
- Row/Col transposition
- Double transposition.

(Substitution)

- Monoalphabetic
 - Polyalphabetic
 - Caesar Cipher
 - Play Fair
 - Hill cipher
 - One time pad
- Vigenere (table)
 → Vennam (One time pad)

1. Monoalphabetic Substitution Cipher :

- A single cipher alphabet for each plain text alphabet is used throughout the process.
- Relation between a character in plain text to a symbol in cipher text is always one to one.

2. Polyalphabetic Substitution Cipher :

- There is no fixed substitution each occurrence of a character may have a different substitute.
- The relationship between a character in plain text to a character in cipher text is one to many.

welcome to SGGs college of - Nanded.
 PRDAYAR IF VEEV AYDORER yv.
 GNGTRT

* Keyless

* Rail Fence :

Plain text is written down as a sequence of diagonals and then read as a sequence of rows.

ex. Meet me at park. (depth $\rightarrow 2$)

M e m a p r
e t e t a k

Cipher text $\rightarrow$ Memaretetak.

ex.2. Help me to Solve the problem (depth $\rightarrow 3$)

X H e p t o o e t e p o b e m
i e s v h x l

✓ H e p m e t o s o i v e t h e p r o b e m

Cipher text : HmSeplepeoortebeltihom.

* Row Column Transformation :

It is more complex than rail fence. we have to write it row by row and read it column by column.

Depends on order of the column.

ex. Tomorrow is class test & Be Ready all

Key (7,5).

key: 4312567

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Date : _____

4	3	1	2	5	6	7
T	O	M	O	R	R	O
W	I	S	C	L	A	S
S	T	E	S	T	B	E
R	E	A	D	Y	A	L
L	U	V	W	X	Y	Z

dummy bits.

Cipher text: MSEAV OCSDW OJT EUTWSRLRLTxxRABAY
OSELZ

Key: ZEBRA
↓ ↓ ↓ ↓ ↓
5 3 2 4 1

STRIKE
↓ ↓ ↓ ↓ ↓
5 5 4 2 3 1

- * Double Transposition:
 - Use cipher text again as Plain text and encrypt again.
 - Key is decided by the alphabetical order of letters given by the sender.
 - Repeat row or column transposition for double transposition.

* Keyed Transposition

Here we divide plain text into groups of predetermined size (blocks) and then use a key to permute the characters in each block separately.

ex. Enemy Attacks Tonight

Block Size $\rightarrow 5$.

Enemy Attac ks Ton ight z $\xleftarrow{\text{dummy}}$

Encryption $\downarrow$

3	1	4	5	2
1	2	3	4	5

$\uparrow$ Decryption

Cipher : eEmyn tAact Tkons hitzg
Text

Ex. Attack postponed tomorrow

Attac kpost poned tomor row yz

E $\downarrow$

4	3	1	2
1	2	3	4

$\uparrow$ D

~~at~~ Attac kpo stpo nedt omor row z
 $\Downarrow$ encryption

atAt opck opst tdne xoom zwxx
 $\Downarrow$ decryption

Atta ckpo stpo nedt omor rowz

* Substitution Techniques:

1. Caesar Cipher.

$$C = E(k, p) = (P + k) \bmod 26$$

$$p = D(k, P) = (C - k) \bmod 26$$

- It is also called as Shift Cipher or additive Cipher
- Each letter in the plain text is replaced by a letter corresponding to the numbers of shifts in the alphabet
- It is monoalphabetic Cipher
- It is one of the earliest and simplest method of Cipher.

If $(C - k)$ is negative then add 26

The key is in between $1 \leq k \leq 25$

A	B	C	D	E	F	G	H
0	1	2	3	4	5	6	7

I	J	K	L	M	N	O	P
8	9	10	11	12	13	14	15

Q	R	S	T	U	V	W	X
16	17	18	19	20	21	22	23

X	Y
24	25

Ex: Plain Text: Hello, key 4.

$$C(H) = (7 + 4) \% 26 = L$$

$$C(e) = (4 + 4) \% 26 = I$$

$$C(L) = (11 + 4) \div 26 = P$$

$$C(L) = (11 + 4) \div 26 = P$$

$$C(O) = (14 + 4) \div 26 = S$$

Hello $\longleftrightarrow$ LIPPS

CT: LIPPS

$$P(L) = (11 - 4) \div 26 = H$$

$$P(I) = (8 - 4) \div 26 = E$$

$$P(P) = (15 - 4) \div 26 = L$$

$$P(S) = (18 - 4) \div 26 = O$$

Ex: Hello key = 15.

$$C(H) = (7 + 15) \div 26 = 22 = W$$

$$C(e) = (4 + 15) \div 26 = 19 = T$$

$$C(l) = (11 + 15) \div 26 = 0 = A$$

$$C(O) = (14 + 15) \div 26 = 3 = D$$

CT: WTAAD

$$P(W) = (22 - 15) \div 26 = 7 = H$$

$$P(T) = (19 - 15) \div 26 = 4 = E$$

$$P(A) = (0 - 15) \div 26 = 11 = L$$

$$P(D) = (3 - 15) \div 26 = 14 = O$$

★ Play Fair cipher algorithm

- It was invented in 1854 by Charles Whitstone but was named after Lord Play Fair who promoted the use of play fair algorithm.

S1. Create 5×5 matrix that is called read of letters.

S2. The matrix is made by inserting the values of key and remaining alphabets into the matrix row wise from left to right where letters I and J will be combined together.

S3. Convert the text into pairs of alphabet

Case 1 • Pair cannot be made with same letters then rate the letters into single and add X to the previous letter.

Hello $\rightarrow$ He IX lo HALLOE $\rightarrow$ HA LX LO EZ

Case 2 • If the letter is standing alone in the process of pairing add Z with the letter.

He x x o e $\rightarrow$ He xz xo ez

Case 3 • Code will be formed using three rules

i) If both alphabets are in same row replace them with alphabets to their immediate right.

ii) If both alphabets are in same column replace them with alphabets immediately below them.

iii) If not in same row or column replace them with alphabets in the same row respectively but add other pair of corners.

Ex Hello.

Keyword : ABITS.

A	B	H	J/I	C
D	E	F	G	K
L	M	N	O	P
Q	R	S	T	U
V	W	X	Y	Z

Help → He Ip (Pairing)
 ↓ ↓
 BF ML

Ex. Tree is green.
 Keyword PLATFAIR

Ex. Hello World
 keyword: MONARCHY

P	L	A	T	F
I /I	R	B	C	D
E	G	H	K	M
N	O	Q	S	T
U	V	W	X	Z

★ Hill Cipher (Polyalphabetic)

• It is developed by Lester Hill in 1929.

• It encrypts a group of letters called polygrapher (diagraph, trigraph etc)

• This method makes use of following to encrypt & decrypt

$$* C = K \cdot P \text{ mod } 26$$

$$* P = K^{-1} \cdot C \text{ mod } 26$$

• First Step is to choose a key
ex. V I E W

$$\begin{bmatrix} V & I \\ E & W \end{bmatrix} = \begin{bmatrix} 21 & 8 \\ 4 & 22 \end{bmatrix}$$

ex. Quickness. (trigraph)

$$\begin{bmatrix} Q & U & I \\ C & K & N \\ E & S & S \end{bmatrix} = \begin{bmatrix} 16 & 20 & 8 \\ 2 & 10 & 13 \\ 4 & 19 & 18 \end{bmatrix}$$

Eq Plain text : ATTACK.

$$\text{key} \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$$

AT TA CK($\because$ key is 2×2)

$$\begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} A \\ T \end{bmatrix} \begin{bmatrix} T \\ A \end{bmatrix} \begin{bmatrix} C \\ K \end{bmatrix}$$

$2 \times 2 \quad 2 \times 1 \quad 2 \times 1 \quad 2 \times 1$

$$C \begin{bmatrix} A \\ T \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 19 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 2 \times 0 + 3 \times 19 \\ 3 \times 0 + 6 \times 19 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 57 \\ 114 \end{bmatrix} \pmod{26} = \begin{bmatrix} 5 \\ 10 \end{bmatrix} = \begin{bmatrix} F \\ K \end{bmatrix}$$

$$C \begin{bmatrix} T \\ A \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 19 \\ 0 \end{bmatrix} \pmod{26} = \begin{bmatrix} 38 \\ 57 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 12 \\ 5 \end{bmatrix} = \begin{bmatrix} M \\ F \end{bmatrix}$$

$$C \begin{bmatrix} C \\ K \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 10 \end{bmatrix} = \begin{bmatrix} 4 + 30 \\ 6 + 60 \end{bmatrix} \pmod{26} = \begin{bmatrix} 34 \\ 66 \end{bmatrix} \pmod{26} = \begin{bmatrix} 8 \\ 14 \end{bmatrix} = \begin{bmatrix} I \\ O \end{bmatrix}$$

ATTACK $\longleftrightarrow$ FKMFI O

$$K^{-1} = \frac{1}{|K|} \text{adj}(K)$$

To decrypt we need K^{-1} value

$$K^{-1} = \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix} \times \frac{1}{3}$$

$$d * d^{-1} = 1 \pmod{26} \quad 26(n)+1$$

$$3 \times 9 = 1 \pmod{26} \Rightarrow \underline{d^{-1} = 9}$$

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$$\text{adj}(K) = \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix}$$

Convert to positive

$$\begin{bmatrix} 6 & 23 \\ 23 & 2 \end{bmatrix} \quad (\text{Add } 26 \text{ to } -ve \text{ digits})$$

$$K^{-1} = \frac{1}{|d|} \text{adj}(K) = d^{-1} \text{adj}(K)$$

$$= 9 \begin{bmatrix} 6 & 23 \\ 23 & 2 \end{bmatrix} = \begin{bmatrix} 54 & 207 \\ 207 & 18 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix}$$

$$P \begin{bmatrix} F \\ K \end{bmatrix} = K^{-1} C \pmod{26}$$

$$= \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 5 \\ 10 \end{bmatrix} \pmod{26} = \begin{bmatrix} 10 + 250 \\ 125 + 180 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 260 \\ 305 \end{bmatrix} = \begin{bmatrix} 0 \\ 19 \end{bmatrix} = \begin{bmatrix} A \\ T \end{bmatrix}$$

$$P \begin{bmatrix} M \\ F \end{bmatrix} = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 12 \\ 5 \end{bmatrix} \pmod{26} = \begin{bmatrix} 24 + 125 \\ 300 + 90 \end{bmatrix} = \begin{bmatrix} 149 \\ 390 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 197 \\ 0 \end{bmatrix} = \begin{bmatrix} T \\ A \end{bmatrix}$$

$$P \begin{bmatrix} J \\ 0 \end{bmatrix} = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 8 \\ 14 \end{bmatrix} \pmod{26} = \begin{bmatrix} 366 \\ 452 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 2 \\ 10 \end{bmatrix} = \begin{bmatrix} C \\ K \end{bmatrix}$$

PT : ATTACK

* Hill Cipher 3×3 matrix.

Ex. plain text - SAFE MESSAGES.

key $\begin{bmatrix} C & I & P \\ H & E & R \\ J & N & G \end{bmatrix}$

SAF EME SSA GES

$$\begin{bmatrix} 2 & 8 & 15 \\ 7 & 4 & 17 \\ 8 & 13 & 6 \end{bmatrix}, \begin{bmatrix} S \\ A \\ F \end{bmatrix} = \begin{bmatrix} 18 \\ 0 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 8 & 15 \\ 7 & 4 & 17 \\ 8 & 13 & 6 \end{bmatrix} \begin{bmatrix} 18 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 111 \\ 211 \\ 174 \end{bmatrix} \pmod{26} = \begin{bmatrix} 7 \\ 3 \\ 18 \end{bmatrix} = \begin{bmatrix} H \\ D \\ S \end{bmatrix}$$

$$\begin{bmatrix} 2 & 8 & 15 \\ 7 & 4 & 17 \\ 8 & 13 & 6 \end{bmatrix} \begin{bmatrix} 4 \\ 12 \\ 4 \end{bmatrix} = \begin{bmatrix} 164 \\ 144 \\ 212 \end{bmatrix} \pmod{26} = \begin{bmatrix} 8 \\ 14 \\ 4 \end{bmatrix} = \begin{bmatrix} I \\ O \\ E \end{bmatrix}$$

$$\begin{bmatrix} 2 & 8 & 15 \\ 7 & 4 & 17 \\ 8 & 13 & 6 \end{bmatrix} \begin{bmatrix} 18 \\ 18 \\ 0 \end{bmatrix} = \begin{bmatrix} 180 \\ 198 \\ 378 \end{bmatrix} \pmod{26} = \begin{bmatrix} 24 \\ 16 \\ 14 \end{bmatrix} = \begin{bmatrix} Q \\ O \\ O \end{bmatrix}$$

$$\begin{bmatrix} 2 & 8 & 15 \\ 7 & 4 & 17 \\ 8 & 13 & 6 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \\ 18 \end{bmatrix} = \begin{bmatrix} 314 \\ 364 \\ 208 \end{bmatrix} \pmod{26} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} C \\ A \\ A \end{bmatrix}$$

SAFE MESSAGES $\Leftrightarrow$ HDSS IOEYQOCAA

$$\begin{aligned} 1D) &= 2(24 - 221) - 8(42 - 136) + 15(91 - 32) \\ &= 2(-197) - 8(-94) + 15(59) \\ &= -394 + 752 + 885 \\ &= 1243 \end{aligned}$$

$$\begin{array}{r} 3 \times 8 \\ 15 \end{array}$$

$$K^{-1} = \begin{bmatrix} 1 & 2 & 8 & 13 \\ 7 & 4 & 17 \\ 8 & 13 & 6 \end{bmatrix}$$

$$d \times d^{-1} = 1 \pmod{26}$$

$$1243 \times 2 \pmod{26}$$

$$1243 \times 5 \pmod{26} = 1$$

$$d^{-1} = 5$$

$$K^{-1} = 5 \begin{bmatrix} -197 & 147 & 59 \\ 147 & -108 & 38 \\ 76 & 71 & -48 \end{bmatrix}^T$$

$$= 5 \begin{bmatrix} -197 & 147 & 76 \\ 94 & 22 & 71 \\ 59 & 38 & -48 \end{bmatrix}$$

$$= 5 \begin{bmatrix} 11 & 147 & 76 \\ 94 & 22 & 71 \\ 59 & 38 & 4 \end{bmatrix} \pmod{26}$$

$$K^{-1} = 5 \begin{bmatrix} 11 & 17 & 24 \\ 16 & 10 & 19 \\ 7 & 12 & 4 \end{bmatrix} = \begin{bmatrix} 55 & 85 & 120 \\ 80 & 50 & 95 \\ 35 & 60 & 20 \end{bmatrix}$$

$$K^{-1} = \begin{bmatrix} 3 & 7 & 16 \\ 2 & 6 & 17 \\ 9 & 18 & 20 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 16 \\ 2 & 6 & 17 \\ 9 & 18 & 20 \end{bmatrix} \begin{bmatrix} 7 \\ 3 \\ 18 \end{bmatrix} = \begin{bmatrix} 18 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} S \\ A \\ F \end{bmatrix}$$

$$2. \begin{bmatrix} 3 & 7 & 16 \\ 2 & 6 & 17 \\ 9 & 18 & 20 \end{bmatrix} \begin{bmatrix} 8 \\ 14 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \\ 4 \end{bmatrix} = \begin{bmatrix} S \\ A \\ E \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 16 \\ 2 & 6 & 17 \\ 9 & 18 & 20 \end{bmatrix} \begin{bmatrix} 24 \\ 16 \\ 14 \end{bmatrix} = \begin{bmatrix} 18 \\ 18 \\ 0 \end{bmatrix} = \begin{bmatrix} S \\ S \\ A \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 16 \\ 2 & 6 & 17 \\ 9 & 18 & 20 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 18 \end{bmatrix} = \begin{bmatrix} G \\ F \\ S \end{bmatrix}$$

Plain text :- SAFE MESSAGE

- ★ Vigenere Cipher : Designed by.
16th Century French Mathematician
- It is poly alphabetic Substitution cipher.
 - Encryption is done using 26×26 matrix

• Method 1 :

	a	b	c	d	e	f	...	z
a	a	b	c	d	e	f	...	z
b	b	c	d	e	f	g	...	A
c	c	d	e	f	g	h	...	B
d							...	
e							...	
f							...	
...							...	
z							...	

Ex. Plain Text : GIVE MONEY

Key : LOCK.

P	G	I	V	E	M	O	N	E	Y	
(11)										
K	L	O	C	K	L	O	C	K	L	
(row)										
C	R	W	X	O	X	C	P	O	J	Cipher
	L	O	C	K	L	O	C	K	L	
Decrypted P.	G	I	V	E	M	O	N	E	Y	Plain text.

G.

← decryption

L → (R)

• Method 2 :

$$E_i = (P_i + k_i) \bmod 26 \quad // \text{Encryption}$$

$$D_i = (C_i - k_i) \bmod 26 \quad // \text{Decryption} \quad \text{-ve} + 26$$

	P	G	I	V	E	M	O	N	E	Y
15	6	8	21	4	12	14	13	4	23	
K	L	O	C	K	L	O	C	K	L	
	11	14	2	10	11	14	20	20	11	
C	17	22	23	14	23	28	15	14	39	
	R	W	X	O	X	C	P	O	J	
C%26	17	22	23	14	23	2	15	14	9	
P	R	W	X	O	X	C	P	O	J	
C - K	6	8	21	4	12	-12	13	10	15	
making +ve (KEX) x 26	6	8	21	4	12	14	13	10	24	
P	G	I	V	E	M	O	N	E	Y	

K: PASCAL.

* Vernam (One time Pad)

20. In this we assign a number same characters in plain text to a number in key
i.e. $\text{length}(\text{key}) = \text{length}(\text{Plain text})$

Ex. Plain Text: RAMSWARUPK

25 Key → RANCHO BABA

P.T.	R	A	M	S	W	A	R	U	P	K.
key	R	A	N	C	H	O	B	A	B	A
PT+key	34	0	-25	20	29	14	18	20	16	10
(PT+key)%26=C	8	0	25	20	3	14	18	20	16	10
C-K.	-9	0	12	18	-4	0	17	20	15	10

Identify the algo (Vegenere/ Verman) and Solve accordingly
 * Verman require $\text{len}(P) = \text{len}(key)$

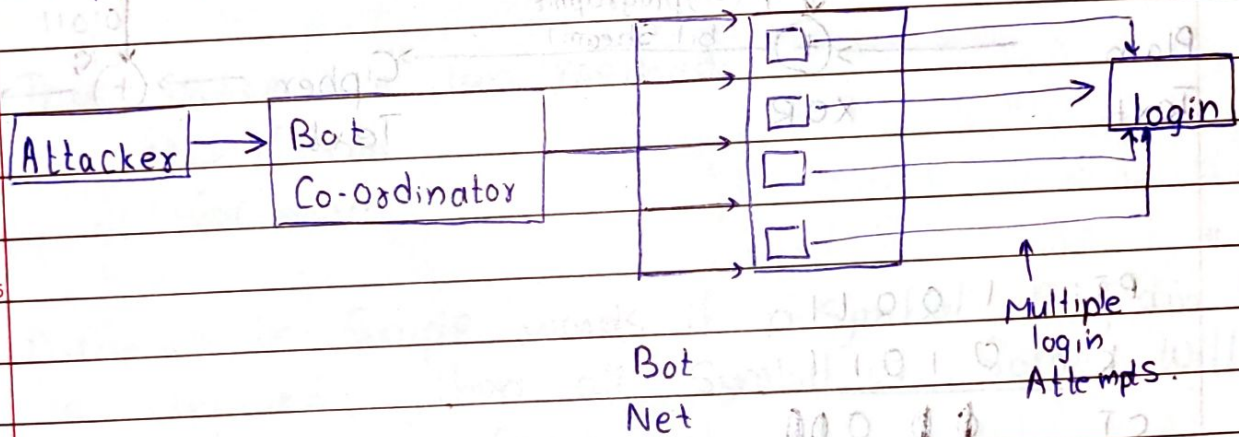
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Date : _____

+ve (C-K)=P	17	0	12	18	22	10	17	20	15	10
	R	A	M	S	W	A	R	U	P	K

* 5 Brute Force Attack:

- It is finding all possible keys and trying all possible key to find the encrypted message.
- Software tools that can perform brute force attack are aircrack-ng, Davegroh, John the Ripper, Hashcat, Hydra, etc.



• One of the way to Stop brute force attack is Captcha to detect whether human being is attacking on a system or not.

• These are different types of captcha.

• Text based Captcha.

• ReCaptcha ☐ I am not a robot.

• 3D Captcha.

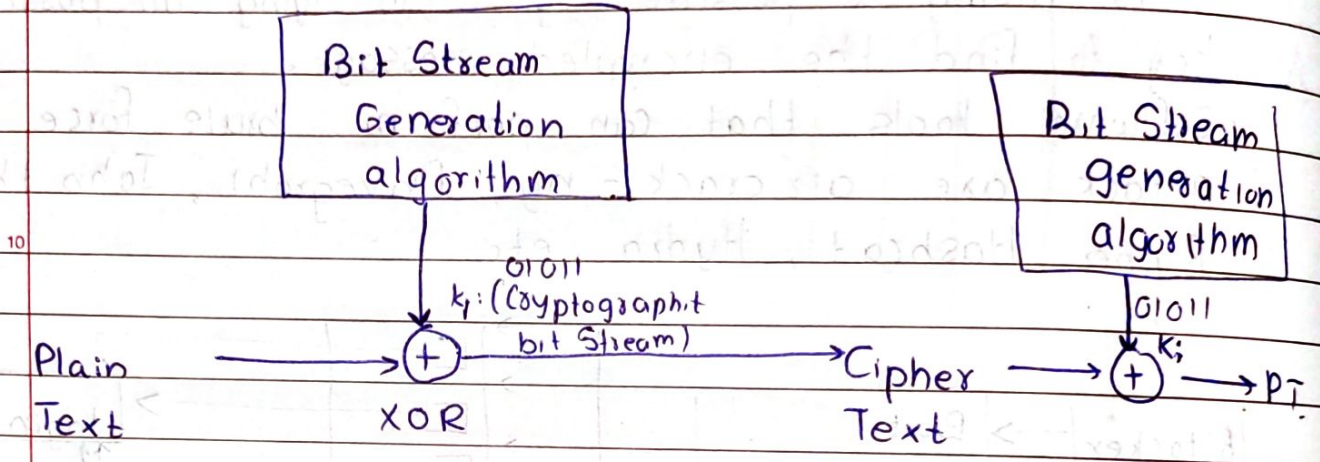
• Mathematical Captcha.

• Image based Captcha

30 Captcha is Completely automated public turing test to tell computers and humans apart.

* Stream & Block Cipher

- * Stream Cipher: It is one that encrypts digital data stream one byte or one bit at a time
- * It is Symmetric key cipher.

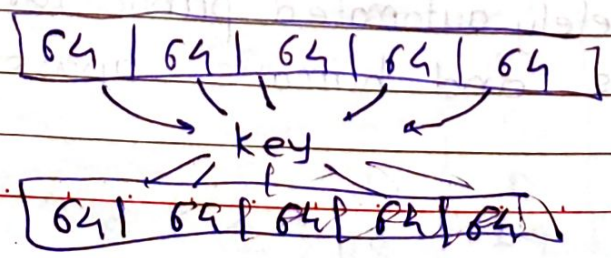


$$\begin{array}{r}
 PT: \quad 10011 \\
 k: \quad 01011 \\
 \hline
 CT: \quad 11000 \\
 k: \quad 01011 \\
 \hline
 PT: \quad 10011
 \end{array}$$

Block Cipher

In this a block of plain text is treated as a whole and used to produce the cipher text of equal length.

- Typically a block size of 64 bit and 128 bits are used
- Key will be applied on each block



* Shanon's Theory of Confusion & Diffusion.

It was introduced by Claude Shanon.

Shanon's concern was to prevent Crypt analysis based on Statistical analysis.

Assume Attacker has Some knowledge of Statistical characters of plain text.

If this statics are in any way reflected in the cipher text. The crypt analyst or attacker may ~~not~~^{be able} to find encryption key.

Thus, there are two methods

1) Diffusion

2) Confusion.

1) Diffusion: In Simple words if a Symbol in plain text is changed then all Symbol or Some will also change in Cipher text.

• The idea of diffusion is to hide the relationship between Cipher text and plain text.

2) Confusion:

• It hides the relationship between cipher text and the key. If a single bit in a key is changed then most or all bits of the cipher text will also change.

• Confusion means each bit of cipher text should depend on several parts of key to make the connection between key and cipher unclear or difficult to understand.

* Difference Between Public and Private key Cryptography

Private key:

- It is only known by Sender and receiver

* Feistel Cipher Structure:

- Most of the ^{block} cipher technique follows this Structure.

Steps:

1) The plain text is divided in two equal parts called as L_0 and R_0 .

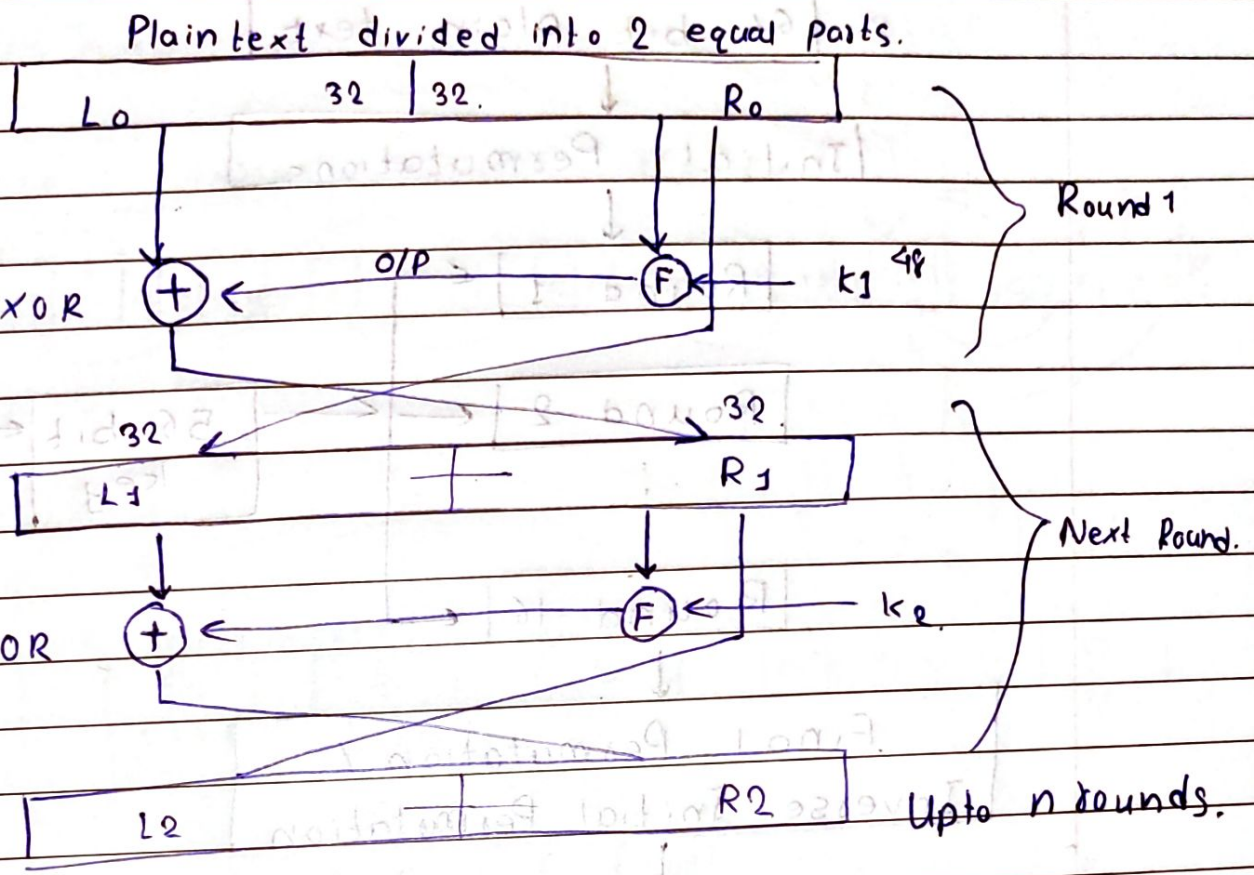
2) The two halves of the data passes through N rounds of processing and then combine to produce the cipher text block.

3) On the right half we apply a function f and then we use subkey generated from master key.

4) The output of this is XORed with the left half and then this output will be Swapped

This is one single round. We will have N rounds depends on algorithm and all rounds will have the same structure.

• In this algorithm we divide the plain text in two equal parts and apply the function on RHS and XOR it with LHS and the output is Swapped then that algorithm follows Feistel Structure.

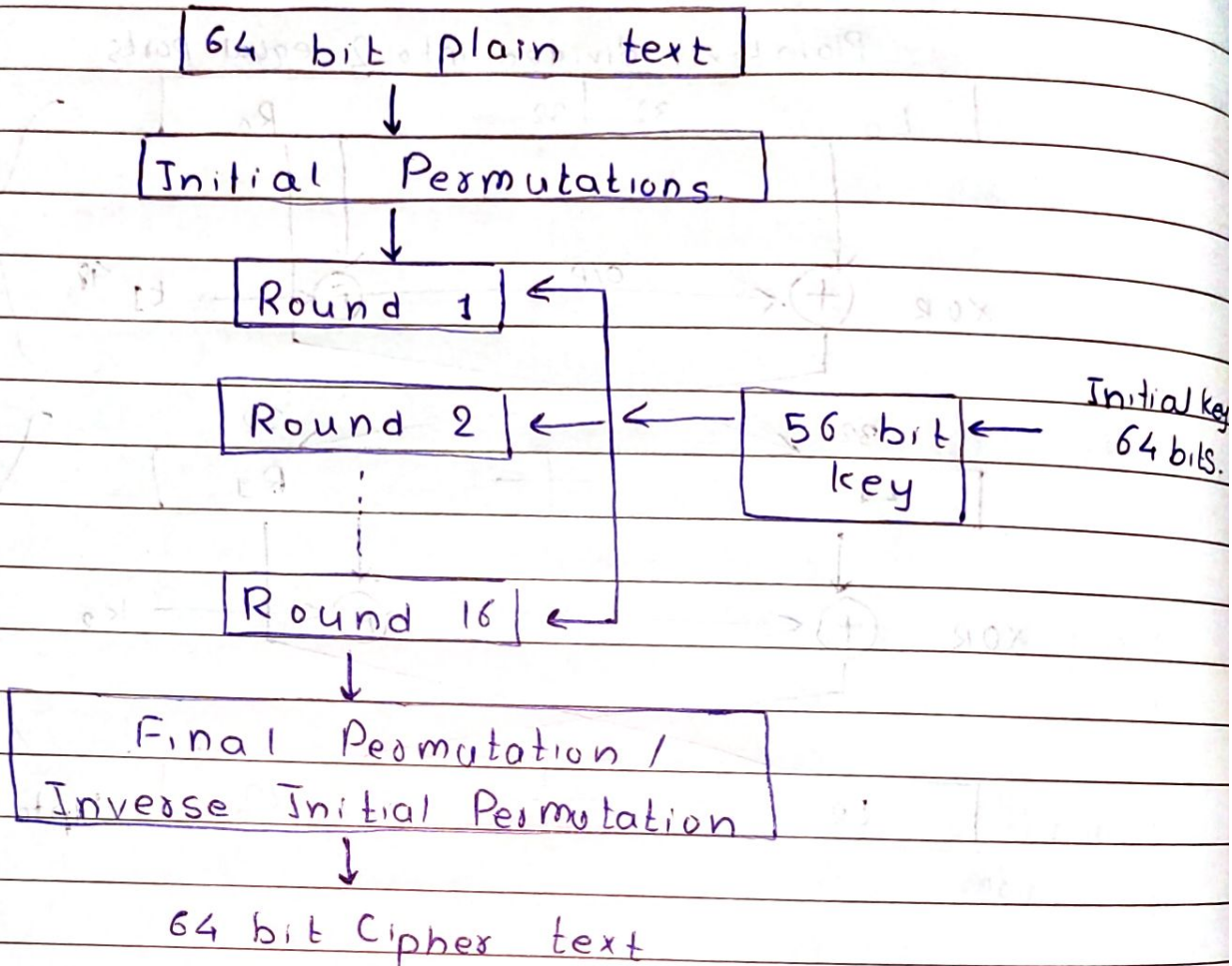


★ DES :

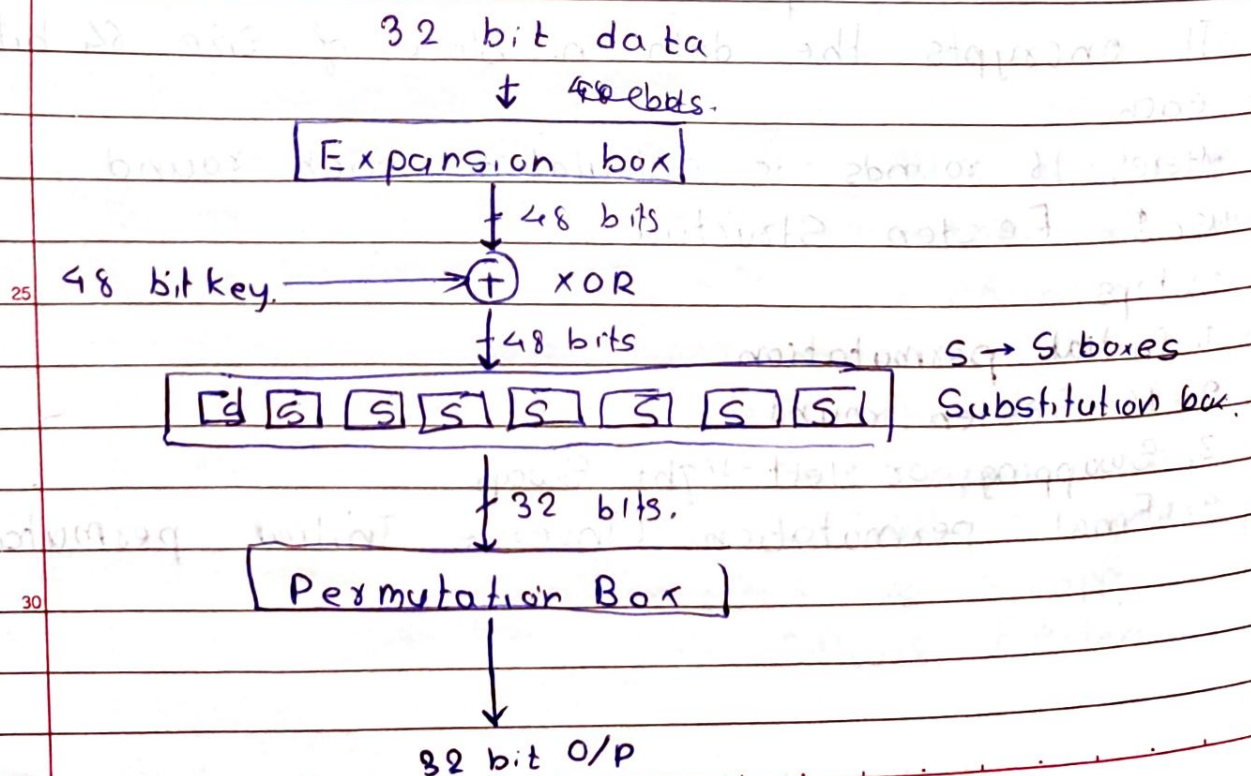
- It is block Cipher
- It is Symmetric technique
- It has 64 bit plain text block
- It encrypts the data in blocks of size 64 bits each
- Here, 16 rounds is calculated, each round uses Feistel Structure.

• Steps :

1. Initial permutation
2. 16 Feistel rounds
3. Swapping or left-right Swap
4. Final permutation (Inverse Initial permutation)

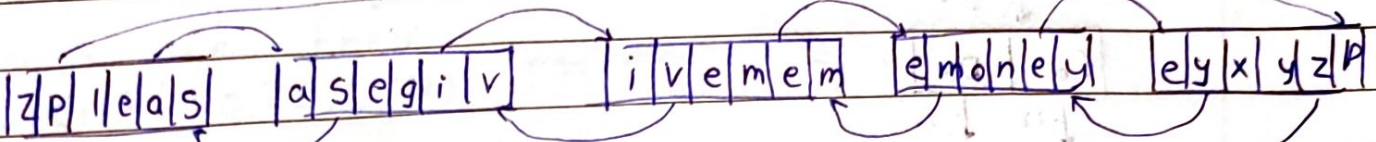


* How Function works :

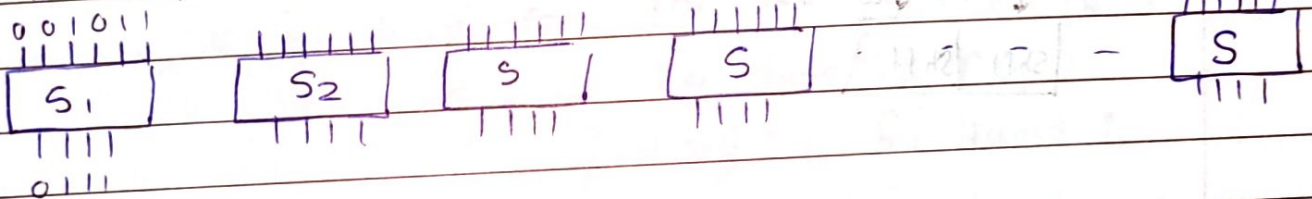


* Expansion Box :

Please give me money.



* 10 S boxes



0101 → 5 (col)
01 → 1 (row)

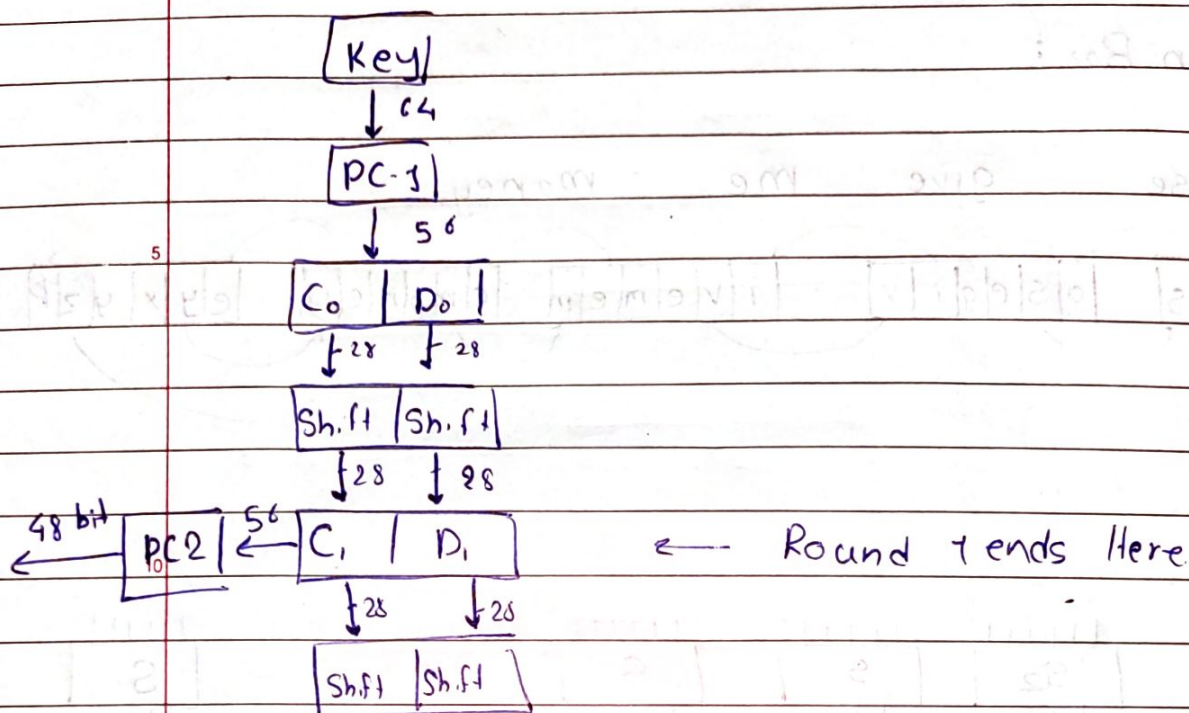
O/p = 7.

0111 → Result.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	3	5	7	0	1	4											
1	4	2	1	0	9	7											

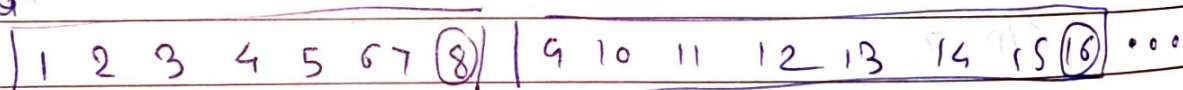
* How 16 Subkeys are generated?

We have 64 bit key which goes as a input to PC1 (Permuted Choice 1) and we get output as 56 bit key



- 15 Inside PC1 (Permuted Choice 1) 64 bit key is divided into 8 parts each of 8 bits. ($8 \times 8 = 64$)

a.



last bit discard.

- From each part last bit is discarded i.e. bit no 8, 16, 24, ..., 24. Hence we get 8 parts of 7 bits each i.e. $7 \times 8 = 56$.
- 25 Output of PC1 is 56 bits which is then divided into two equal parts 28 bits each i.e. C_0 and D_0 now that bits are shifted left in each round.
- In rounds $i = 1, 2, 9, 16$ 1 Shift rotated left.
- 30 In other rounds, two shift rotated towards left.

- Inside PC2 56 bits are converted to 48 bits using pre defined table
- In C_1 28 bits are there and in D_1 also 28 bits are there.
 $C_1 = (1-28)$, $D_1 = (29-56)$.
- Now 56 bits $\rightarrow$ 48 bits
- Next half C_1 (9, 8, 22, 25) position bits are removed or discarded i.e. now C_1 is of 24 bits
- Similarly Right half D_1 (35, 38, 33, 54) position bits are discarded leaving 24 bits in D_1 .
- Then we get over our first key K_1 for round 1.

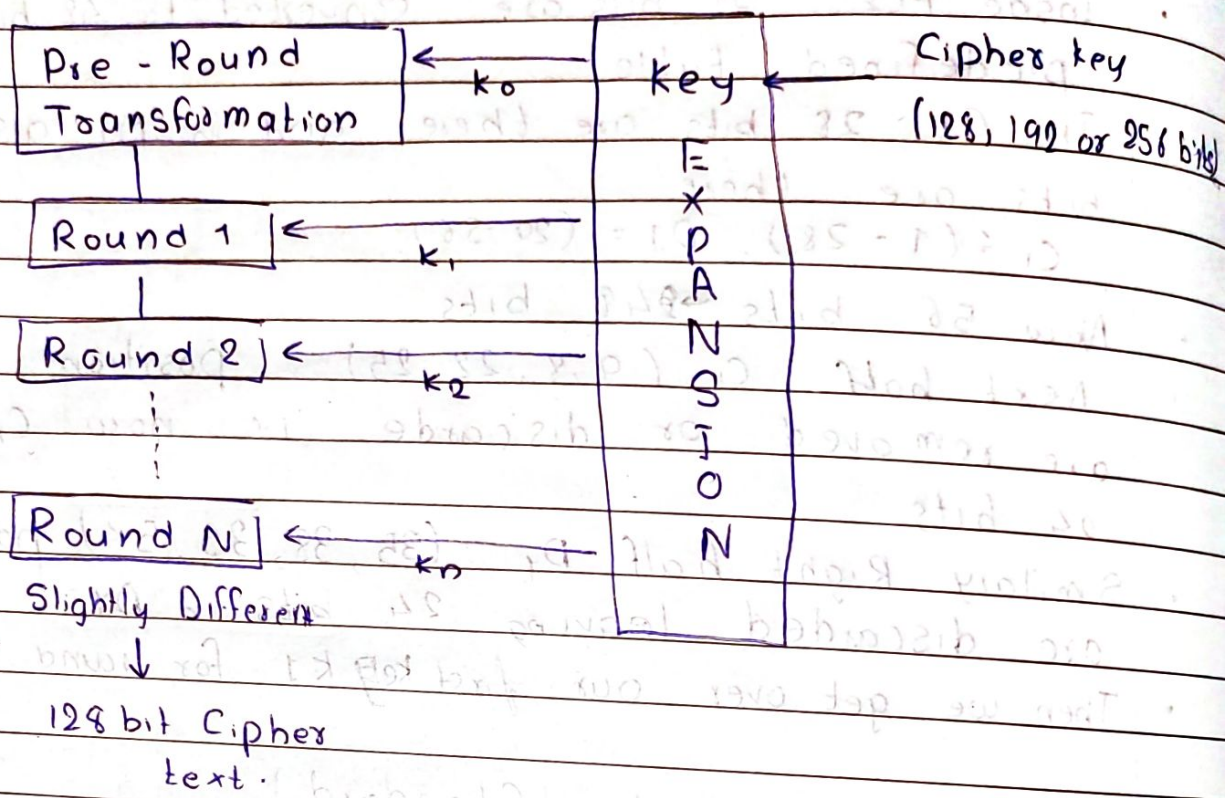
★ AES (Advanced Encryption Standard)

- It is Symmetric key block cipher
- Established in 2001 by NIST.
- It has fixed block size equals to 128 bits i.e. 16 bytes - equal to 4 words.

Rounds	No of key used.	
10	128 bits	AES 128 Version
12	192	AES 192 Version
14	256	AES 256 Version

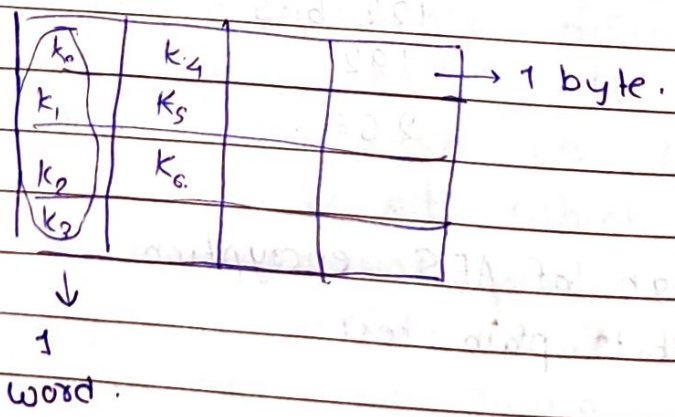
- ★ General design of AES encryption
 128 bits plain text

128 - bits plain text.



• No of keys generated by key expansion algorithm equals to no of round + 1.

State = 16 bytes (4×4) : It stores intermediate results.



State Array:

$S_{0,0}$	$S_{0,1}$	$S_{0,2}$	$S_{0,3}$
$S_{1,0}$	$S_{1,1}$	$S_{1,2}$	$S_{1,3}$
$S_{2,0}$	$S_{2,1}$	$S_{2,2}$	$S_{2,3}$
$S_{3,0}$	$S_{3,1}$	$S_{3,2}$	$S_{3,3}$

★ Structure of each round:
Plain text (128 bits)

Expand key

Only
XOR

Add Round key

$(W_0 - W_3)$

Substitute Bytes

Shift Rows

Mix Column

Add round key

$(W_4 - W_7)$

Substitute Bytes

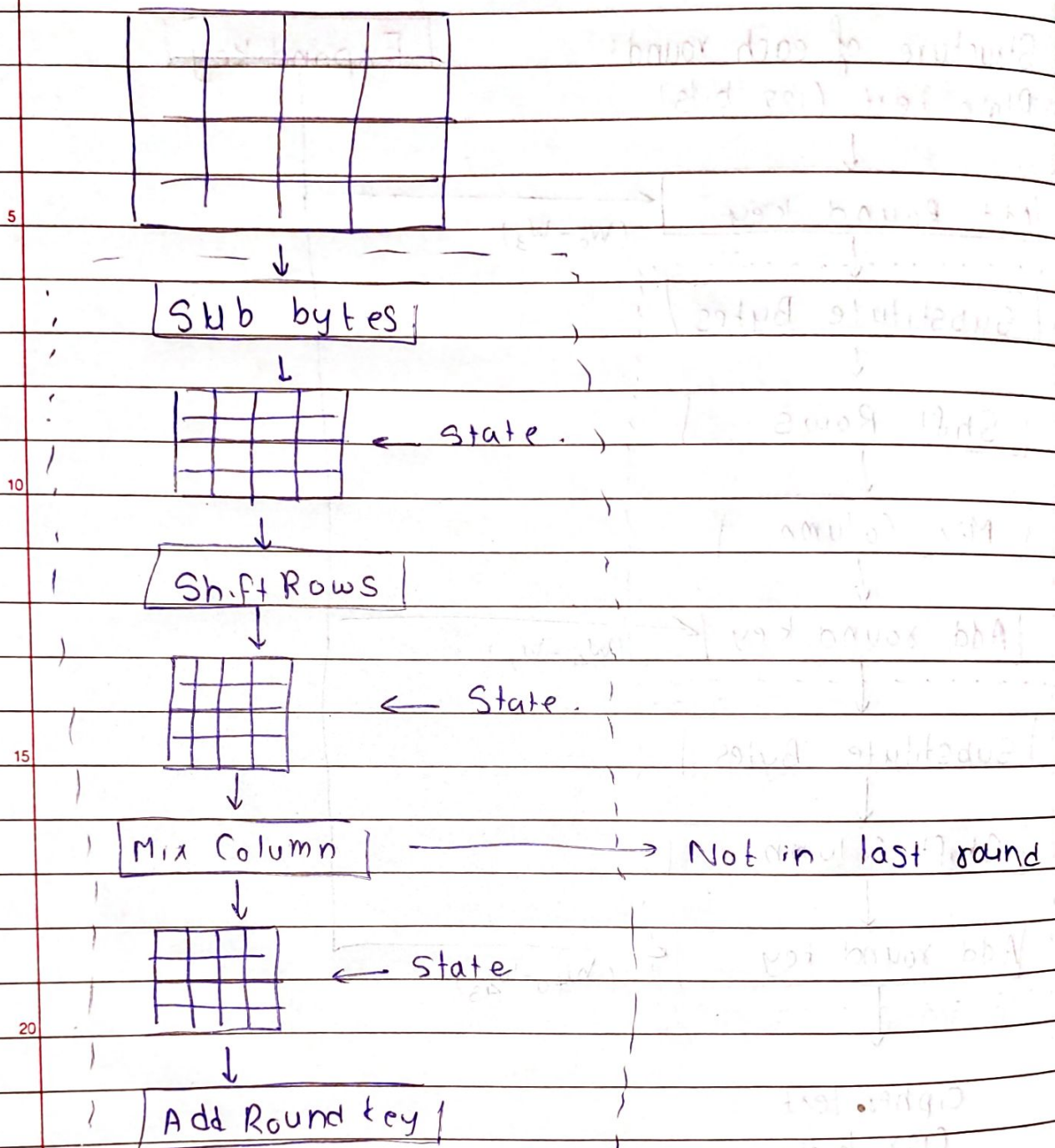
Shift Column

Add round key

$(W_{40} - W_{43})$

Cipher text
(128 bits)

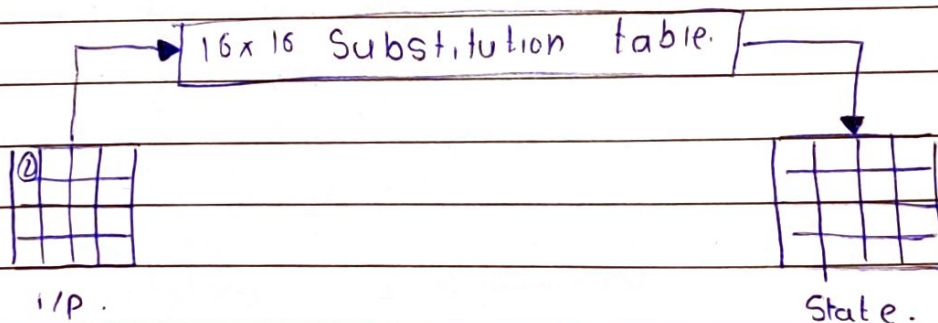
- Each round except the last round uses four transformation
- In last round mixed column transformation is missing.



There are four types of transformation

- Substitution
- Permutation
- Mixing
- Key adding

- Like DES, AES also does Substitution but mechanism is different.
- Substitution here is done on each byte
 - Only one table is used for transformation of bits
 - Which means that if 2 bytes are same then transformation is also same.
- Here Hexadecimal digits are used
 - first hexadecimal digit determines row
 - Second hexadecimal digit determines column of the substitution table.



② → 0000 0010

→ 0 2
 ↓ ↓
 row column

	0	1	2	3	4	...	16
0	0	1	2	1	3		
1	1	3	2	1			
2	3	1	1	3	5		
...	1	5	7	6			
16							